

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1, CO2

Answer the following:

- (a) Define the Coriolis acceleration and state when it becomes significant in engineering analysis. [0.5]

Ans:

The Coriolis acceleration is the term $2\Omega \times v_{rel}$ that arises when an object moves with velocity v_{rel} in a reference frame rotating at angular velocity Ω . It becomes significant when the product $\Omega \cdot v_{rel}$ is comparable to other acceleration terms, e.g. in large-scale meteorology, long-range ballistics, or fast-rotating machinery.

- (b) What is an Atwood machine? State the underlying principle on which it operates. [0.5]

Ans:

An Atwood machine consists of two masses connected by a light inextensible string passing over a frictionless pulley. It works on Newton's second law applied to a constrained two-body system: the heavier mass accelerates downward while the lighter mass accelerates upward, with a common acceleration determined by the mass difference.

- (c) Define the moment of inertia of a rigid body and state its SI unit. [0.5]

Ans:

The moment of inertia (I) of a rigid body about a given axis is the sum (or integral) of the products of each mass element and the square of its distance from the axis: $I = \sum m_i r_i^2$ (discrete) or $I = \int r^2 dm$ (continuous). SI unit: $\text{kg} \cdot \text{m}^2$.

- (d) Differentiate between undamped and damped free vibrations. [0.5]

Ans:

Undamped free vibration: the system oscillates indefinitely at its natural frequency ω_n with constant amplitude (no energy loss). Damped free vibration: a dissipative element (damper, friction) causes the amplitude to decay over time; the oscillation frequency ω_d is slightly lower than ω_n , and the system eventually comes to rest.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A solid disk of mass 5 kg and radius 0.2 m rolls without slipping. Calculate its moment of inertia about the contact point using the parallel axis theorem. [0.5]

Ans:

$I_{cm} = (1/2)mR^2 = 0.5 \times 5 \times 0.2^2 = 0.10 \text{ kg} \cdot \text{m}^2$. By parallel axis theorem: $I_{contact} = I_{cm} + mR^2 = 0.10 + 5 \times 0.04 = 0.10 + 0.20 = 0.30 \text{ kg} \cdot \text{m}^2$.

- (b) Explain why a solid sphere reaches the bottom of an incline before a hollow cylinder of the same mass and radius. [0.5]

Ans:

The rolling acceleration is $a = g \sin\theta / (1 + \beta)$, where $\beta = I/(mR^2)$. Solid sphere: $\beta = 2/5 = 0.4$, hollow cylinder: $\beta = 1.0$. Since the sphere has a smaller β , it has a larger acceleration and reaches the bottom first. The hollow cylinder puts more energy into rotation.

- (c) In an Atwood machine with masses 5 kg and 3 kg, state which mass accelerates downward and why. [0.5]

Ans:

The 5 kg mass accelerates downward because it is heavier. In an Atwood machine, the net downward force is $(m_1 - m_2)g = (5 - 3) \times 9.81 = 19.62 \text{ N}$, and this causes the heavier mass to descend while the lighter mass rises.

- (d) State two assumptions commonly made when analyzing an ideal pulley system. [0.5]

Ans:

(1) The pulley is massless and frictionless (no rotational inertia, no axle friction). (2) The string/rope is massless, inextensible, and has uniform tension throughout its length.

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO2

Two masses $m_1 = 5 \text{ kg}$ and $m_2 = 3 \text{ kg}$ are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81 \text{ m/s}^2$.

- (a) Draw the free-body diagram for each mass and write the equation of motion for m_1 . [0.5]

Ans:

FBD for m_1 : Weight $m_1g = 5 \times 9.81 = 49.05$ N downward, Tension T upward.

Equation of motion (taking downward positive for m_1): $m_1g - T = m_1a$

$$\rightarrow 49.05 - T = 5a$$

- (b) Write the equation of motion for m_2 . [0.5]

Ans:

FBD for m_2 : Weight $m_2g = 3 \times 9.81 = 29.43$ N downward, Tension T upward.

Equation of motion (taking upward positive for m_2): $T - m_2g = m_2a$

$$\rightarrow T - 29.43 = 3a$$

- (c) Find the acceleration of the system. [0.5]

Ans:

Adding both equations: $m_1g - m_2g = (m_1 + m_2)a$

$$a = (m_1 - m_2)g / (m_1 + m_2) = (5 - 3) \times 9.81 / (5 + 3)$$

$$a = 19.62 / 8 = 2.453 \text{ m/s}^2$$

- (d) Calculate the tension in the string. [0.5]

Ans:

From equation for m_2 : $T = m_2(g + a) = 3 \times (9.81 + 2.453)$

$$T = 3 \times 12.263 = 36.79 \text{ N}$$

Check via m_1 : $T = m_1(g - a) = 5 \times 7.357 = 36.79 \text{ N}$ □

- (e) Find the velocity of the masses after 2 seconds. [0.5]

Ans:

$$v = u + at = 0 + 2.453 \times 2 = 4.905 \text{ m/s}$$

- (f) Calculate the distance traveled by each mass in 2 seconds. [0.5]

Ans:

$$s = \frac{1}{2}at^2 = 0.5 \times 2.453 \times 2^2 = 0.5 \times 2.453 \times 4 = 4.905 \text{ m}$$

Q3 [3 Marks]

Bloom: BL 3-4 (Apply / Analyze) | CO2

A platform rotates at constant angular velocity $\Omega = 3$ rad/s about a vertical axis. An object at $r = 2$ m from the center moves radially outward at $v_{\text{rel}} = 1.5$ m/s with no radial acceleration ($\dot{r} = 0$). The angular acceleration of the platform is zero ($\alpha = 0$).

- (a) Calculate the centripetal acceleration magnitude. [0.5]

Ans:

$$a_{\text{centripetal}} = \Omega^2 r = 3.0^2 \times 2.0 = 18.0 \text{ m/s}^2$$

- (b) Calculate the Coriolis acceleration magnitude. [0.5]

Ans:

$$a_{\text{Coriolis}} = 2\Omega v_{\text{rel}} = 2 \times 3.0 \times 1.5 = 9.0 \text{ m/s}^2$$

- (c) State the direction of the Coriolis acceleration relative to the radial velocity. [0.5]

Ans:

The Coriolis acceleration is perpendicular to v_{rel} (the radial direction), directed tangentially in the sense opposite to the platform rotation (by the right-hand rule for $2\Omega \times v_{\text{rel}}$).

- (d) Find the total acceleration magnitude in the rotating frame. [0.5]

Ans:

$$|a_{\text{total}}| = \sqrt{a_{\text{cent}}^2 + a_{\text{Cor}}^2} = \sqrt{(18.0^2 + 9.0^2)} = \sqrt{(324 + 81)} = \sqrt{405} = 20.12 \text{ m/s}^2$$

- (e) Determine the tangential velocity of the object as observed from the fixed (inertial) frame. [0.5]

Ans:

$$v_{\text{tangential}} = \Omega r = 3.0 \times 2.0 = 6.0 \text{ m/s}$$

- (f) If the object continues at the same radial speed, find the centripetal acceleration when $r = 4$ m. [0.5]

Ans:

$$\text{At } r = 4 \text{ m: } a_{\text{centripetal}} = \Omega^2 r = 3.0^2 \times 4 = 36.0 \text{ m/s}^2$$

Q4 [3 Marks]

Bloom: BL 4-5 (Analyze / Evaluate) | CO2, CO3

A uniform rod of mass $m = 2$ kg and length $L = 1$ m is pivoted at one end and oscillates as a physical pendulum under gravity. Take $g = 9.81$ m/s².

- (a) Calculate the moment of inertia of the rod about the pivot using the parallel axis theorem. [0.5]

Ans:

$$I_{\text{cm}} = (1/12)mL^2 = (1/12) \times 2.0 \times 1.0^2 = 0.1667 \text{ kg}\cdot\text{m}^2$$

$$I_{\text{pivot}} = I_{\text{cm}} + m(L/2)^2 = 0.1667 + 2.0 \times 0.25 = 0.1667 + 0.5000 = 0.6667 \text{ kg}\cdot\text{m}^2$$

$$\text{(Or directly: } I_{\text{pivot}} = (1/3)mL^2 = 0.6667 \text{ kg}\cdot\text{m}^2)$$

- (b) Find the distance of the center of mass from the pivot. **[0.5]**

Ans:

$$d = L/2 = 1.0/2 = 0.5 \text{ m}$$

- (c) Write the equation of motion for small-angle oscillations. **[0.5]**

Ans:

$$\text{Restoring torque: } \tau = -mgd \sin\theta \approx -mgd\theta \text{ (small angle)}$$

$$\text{Equation: } I_{\text{pivot}} \ddot{\theta} + mgd \theta = 0 \rightarrow \ddot{\theta} + (mgd/I_{\text{pivot}}) \theta = 0$$

- (d) Determine the natural frequency ω_n . **[0.5]**

Ans:

$$\omega_n = \sqrt{(mgd / I_{\text{pivot}})} = \sqrt{(2.0 \times 9.81 \times 0.5 / 0.6667)}$$

$$= \sqrt{(9.810 / 0.6667)} = \sqrt{14.715} = 3.836 \text{ rad/s}$$

- (e) Calculate the time period of oscillation. **[0.5]**

Ans:

$$T = 2\pi / \omega_n = 2\pi / 3.836 = 1.638 \text{ s}$$

- (f) Compare this period with that of a simple pendulum of the same length ($L = 1 \text{ m}$). **[0.5]**

Ans:

$$\text{Simple pendulum (same length } L = 1.0 \text{ m): } T_{\text{simple}} = 2\pi\sqrt{(L/g)} = 2\pi\sqrt{(1.0/9.81)} = 2.006 \text{ s.}$$

Physical pendulum period (1.638 s) < Simple pendulum period (2.006 s) because the physical pendulum's mass is distributed closer to the pivot, reducing the effective pendulum length.

— End of Set 1 Model Answers —

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1, CO2

Answer the following:

- (a) State the parallel axis theorem and give its mathematical expression. [0.5]

Ans:

The parallel axis theorem states that the moment of inertia about any axis parallel to an axis through the center of mass is: $I = I_{cm} + md^2$, where I_{cm} = MOI about the CM axis, m = total mass, d = perpendicular distance between the two axes. This is the minimum-shift property: I is always smallest about the CM.

- (b) Define the term 'gear ratio' and explain its physical significance. [0.5]

Ans:

The gear ratio (GR) is the ratio of the number of teeth (or radius) of the driven gear to the driving gear: $GR = N_{driven} / N_{driver}$. Physically, it determines the speed and torque transformation: output speed = input speed / GR, and output torque = input torque $\times$ GR (ignoring losses). A $GR > 1$ reduces speed but amplifies torque.

- (c) What is a slider-crank mechanism? Name two practical applications. [0.5]

Ans:

A slider-crank mechanism converts rotary motion (crank) into reciprocating linear motion (slider/piston) via a connecting rod. The kinematic constraint is $r \sin \theta = L \sin \phi$, relating the crank angle θ to the connecting rod angle ϕ . Applications: (1) Internal combustion engines (piston–crankshaft), (2) Reciprocating compressors / pumps.

- (d) Define critical damping and state the mathematical condition for it. [0.5]

Ans:

Critical damping is the boundary between oscillatory (underdamped) and non-oscillatory (overdamped) response. Mathematical condition: damping ratio $\zeta = 1$, i.e. $c = c_{cr} = 2\sqrt{km}$, where c = damping coefficient, k = stiffness, m = mass. The system returns to equilibrium in the shortest time without overshooting.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A gear train has an input speed of 100 RPM and a gear ratio of 15 : 1 (speed reduction). Calculate the output shaft speed. [0.5]

Ans:

Output speed = Input speed / GR = $100 / 15 = 6.67$ RPM.

- (b) Explain the physical meaning of the damping ratio ζ and what happens when $\zeta = 1$. [0.5]

Ans:

The damping ratio $\zeta = c / c_{cr}$ is the ratio of actual damping to critical damping. It determines the system's response type: $\zeta < 1 \rightarrow$ underdamped (oscillates with decaying amplitude), $\zeta = 1 \rightarrow$ critically damped (fastest return to equilibrium without overshoot), $\zeta > 1 \rightarrow$ overdamped (slow, non-oscillatory return).

- (c) A CNC axis must traverse 200 mm with $v_{max} = 100$ mm/s and $a_{max} = 500$ mm/s². Estimate the total traverse time using a trapezoidal profile. [0.5]

Ans:

$t_{accel} = v_{max} / a_{max} = 100/500 = 0.2$ s. $d_{accel} = v_{max}^2 / (2 a_{max}) = 10000/1000 = 10$ mm. $d_{cruise} = 200 - 2 \times 10 = 180$ mm. $t_{cruise} = 180/100 = 1.8$ s. Total time = $0.2 + 1.8 + 0.2 = 2.2$ s.

- (d) State the condition for a block to remain stationary on an inclined plane with friction. [0.5]

Ans:

The block remains stationary when the component of gravity along the incline does not exceed the maximum static friction: $mg \sin \theta \leq \mu_s mg \cos \theta$, i.e. $\tan \theta \leq \mu_s$, where μ_s is the coefficient of static friction.

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO2

Two masses $m_1 = 8$ kg and $m_2 = 5$ kg are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s².

- (a) Draw the free-body diagram for each mass and write the equation of motion for m_1 . [0.5]

Ans:

FBD for m_1 : Weight $m_1 g = 8 \times 9.81 = 78.48$ N downward, Tension T upward.

Equation of motion (taking downward positive for m_1): $m_1g - T = m_1a$

$$\rightarrow 78.48 - T = 8a$$

- (b) Write the equation of motion for m_2 . [0.5]

Ans:

FBD for m_2 : Weight $m_2g = 5 \times 9.81 = 49.05$ N downward, Tension T upward.

Equation of motion (taking upward positive for m_2): $T - m_2g = m_2a$

$$\rightarrow T - 49.05 = 5a$$

- (c) Find the acceleration of the system. [0.5]

Ans:

Adding both equations: $m_1g - m_2g = (m_1 + m_2)a$

$$a = (m_1 - m_2)g / (m_1 + m_2) = (8 - 5) \times 9.81 / (8 + 5)$$

$$a = 29.43 / 13 = 2.264 \text{ m/s}^2$$

- (d) Calculate the tension in the string. [0.5]

Ans:

From equation for m_2 : $T = m_2(g + a) = 5 \times (9.81 + 2.264)$

$$T = 5 \times 12.074 = 60.37 \text{ N}$$

Check via m_1 : $T = m_1(g - a) = 8 \times 7.546 = 60.37 \text{ N} \quad \square$

- (e) Find the velocity of the masses after 3 seconds. [0.5]

Ans:

$$v = u + at = 0 + 2.264 \times 3 = 6.792 \text{ m/s}$$

- (f) Calculate the distance traveled by each mass in 3 seconds. [0.5]

Ans:

$$s = \frac{1}{2}at^2 = 0.5 \times 2.264 \times 3^2 = 0.5 \times 2.264 \times 9 = 10.187 \text{ m}$$

Q3 [3 Marks]

Bloom: BL 3-4 (Apply / Analyze) | CO2

A block of mass $m = 5$ kg is placed on a rough inclined plane at angle $\theta = 30^\circ$ to the horizontal. The coefficient of kinetic friction is $\mu_k = 0.2$. The block is released from rest. Take $g = 9.81 \text{ m/s}^2$.

- (a) Draw the free-body diagram showing all forces on the block. [0.5]

Ans:

Forces: (1) Weight $W = mg = 49.05$ N downward, (2) Normal force N perpendicular to incline (upward from surface), (3) Kinetic friction $f = \mu_k N$ opposing motion (up the incline).

- (b) Calculate the normal reaction force N . [0.5]

Ans:

$$N = mg \cos \theta = 5.0 \times 9.81 \times \cos 30^\circ = 49.05 \times 0.8660 = 42.48 \text{ N}$$

- (c) Calculate the kinetic friction force f . [0.5]

Ans:

$$f = \mu_k \times N = 0.2 \times 42.48 = 8.50 \text{ N}$$

- (d) Determine the net force along the incline and find the acceleration. [0.5]

Ans:

$$F_{\text{net}} = mg \sin \theta - f = 24.52 - 8.50 = 16.03 \text{ N}$$

$$a = F_{\text{net}} / m = 16.03 / 5.0 = 3.21 \text{ m/s}^2$$

- (e) Find the velocity of the block after sliding 5 m from rest. [0.5]

Ans:

$$v = \sqrt{2as} = \sqrt{2 \times 3.21 \times 5.0} = \sqrt{32.06} = 5.66 \text{ m/s}$$

- (f) Calculate the time taken to slide 5 m from rest. [0.5]

Ans:

$$t = \sqrt{2s/a} = \sqrt{2 \times 5.0 / 3.21} = \sqrt{3.12} = 1.77 \text{ s}$$

Q4 [3 Marks]

Bloom: BL 4-5 (Analyze / Evaluate) | CO2, CO3

A mass $m = 2$ kg is attached to a spring of stiffness $k = 200 \text{ N/m}$. The system is displaced $x_0 = 0.1$ m from equilibrium and released from rest ($v_0 = 0$).

- (a) Write the equation of motion for the undamped system. [0.5]

Ans:

$$\text{Equation of motion: } m\ddot{x} + kx = 0 \rightarrow 2.0\ddot{x} + 200.0x = 0$$

- (b) Calculate the natural frequency ω_n and the time period T . [0.5]

Ans:

$$\omega_n = \sqrt{k/m} = \sqrt{(200.0/2.0)} = \sqrt{100} = 10.00 \text{ rad/s}$$

$$T = 2\pi/\omega_n = 2\pi/10.00 = 0.6283 \text{ s} \approx 0.628 \text{ s}$$

- (c) Write the displacement as a function of time $x(t)$ for the given initial conditions. **[0.5]**

Ans:

With $x(0) = 0.1 \text{ m}$, $\dot{x}(0) = 0$:

$$x(t) = x_0 \cos(\omega_n t) = 0.1 \cos(10t) \text{ m}$$

(Amplitude 0.1 m, purely cosinusoidal, no decay)

- (d) A viscous damper with $c = 4 \text{ N}\cdot\text{s/m}$ is now added. Calculate the critical damping coefficient c_{cr} and the damping ratio ζ . **[0.5]**

Ans:

$$c_{cr} = 2\sqrt{km} = 2\sqrt{(200.0 \times 2.0)} = 2\sqrt{400} = 2 \times 20.0 = 40.0 \text{ N}\cdot\text{s/m}$$

$$\zeta = c / c_{cr} = 4.0 / 40.0 = 0.10$$

- (e) State whether the damped system is underdamped, critically damped, or overdamped. Calculate the damped natural frequency ω_d . **[0.5]**

Ans:

Since $\zeta = 0.10 < 1 \rightarrow$ Underdamped (oscillatory with decay).

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 10.00 \sqrt{1 - 0.1^2} = 10.00 \times 0.9950 = 9.95 \text{ rad/s}$$

- (f) Calculate the logarithmic decrement δ for the damped system. **[0.5]**

Ans:

$$\delta = 2\pi\zeta / \sqrt{1 - \zeta^2} = 2\pi \times 0.1 / \sqrt{1 - 0.1^2}$$

$$= 0.6283 / 0.9950 = 0.6315$$

This means each successive peak is $e^{(-\delta)} = e^{(-0.6315)} = 0.5318$ times the previous peak.

— End of Set 2 Model Answers —

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1, CO2

Answer the following:

- (a) Explain the concept of rolling without slipping and state the velocity constraint. [0.5]

Ans:

Rolling without slipping means the contact point of the rolling body has zero velocity relative to the surface. The velocity constraint is $v_{cm} = \omega R$, where v_{cm} = linear velocity of the center of mass, ω = angular velocity, R = radius. Equivalently, $a_{cm} = \alpha R$ for accelerations.

- (b) What is a trapezoidal velocity profile in CNC motion planning? Sketch it qualitatively. [0.5]

Ans:

A trapezoidal velocity profile has three phases: (1) constant acceleration (velocity ramps up linearly), (2) constant velocity (cruise phase), and (3) constant deceleration (velocity ramps down). The velocity vs. time graph forms a trapezoid shape. It is used in CNC and servo drives because it limits the maximum acceleration (jerk control) while achieving a desired feedrate.

- (c) Define logarithmic decrement in vibration analysis and state its physical meaning. [0.5]

Ans:

The logarithmic decrement δ is the natural log of the ratio of two successive peak amplitudes in a damped oscillation: $\delta = \ln(x_n / x_{n+1})$. It is related to the damping ratio by $\delta = 2\pi\zeta / \sqrt{1 - \zeta^2}$. Physically, it quantifies the rate of amplitude decay per cycle — a larger δ means faster energy dissipation.

- (d) State the relationship between mechanical advantage and the number of supporting rope segments in a block-and-tackle system. [0.5]

Ans:

In an ideal block-and-tackle system, the mechanical advantage (MA) equals the number of rope segments supporting the load: $MA = n$. This means the required pulling force is $F = W/n$, but the rope must be pulled n times the load displacement (energy conservation). In a real system, friction reduces the effective MA.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A platform rotates at $\Omega = 3$ rad/s. A particle moves radially outward at $v_{rel} = 2$ m/s. Calculate the magnitude of the Coriolis acceleration. [0.5]

Ans:

$a_{Coriolis} = 2\Omega v_{rel} = 2 \times 3 \times 2 = 12$ m/s². Direction: perpendicular to v_{rel} , opposing the sense of rotation.

- (b) Write the equation of motion for an undamped spring-mass system and identify each term. [0.5]

Ans:

Equation: $m\ddot{x} + kx = 0$, where m = mass (kg), $\ddot{x} = d^2x/dt^2$ = acceleration (m/s²), k = spring stiffness (N/m), x = displacement from equilibrium (m). Term $m\ddot{x}$ = inertia force; term kx = restoring spring force.

- (c) Define mechanical advantage of a pulley system. State its ideal value for a single movable pulley. [0.5]

Ans:

Mechanical advantage (MA) = Load / Effort = W / F . For an ideal single movable pulley, $MA = 2$, since two rope segments support the load. This means the effort is halved, but the rope must be pulled twice the load displacement.

- (d) Explain why fictitious forces (pseudo-forces) appear in a rotating (non-inertial) reference frame. [0.5]

Ans:

Newton's laws are valid only in inertial (non-accelerating) frames. When we write $F = ma$ in a rotating frame, extra terms appear because the frame itself is accelerating. These terms (centrifugal force = $-m\Omega \times (\Omega \times r)$, Coriolis force = $-2m\Omega \times v_{rel}$) are called fictitious or pseudo-forces because they have no physical agent — they arise purely from the non-inertial reference frame.

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO2

Two masses $m_1 = 6$ kg and $m_2 = 4$ kg are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s².

- (a) Draw the free-body diagram for each mass and write the equation of motion for m_1 . [0.5]

Ans:

FBD for m_1 : Weight $m_1g = 6 \times 9.81 = 58.86$ N downward, Tension T upward.

Equation of motion (taking downward positive for m_1): $m_1g - T = m_1a$

$$\rightarrow 58.86 - T = 6a$$

- (b) Write the equation of motion for m_2 . [0.5]

Ans:

FBD for m_2 : Weight $m_2g = 4 \times 9.81 = 39.24$ N downward, Tension T upward.

Equation of motion (taking upward positive for m_2): $T - m_2g = m_2a$

$$\rightarrow T - 39.24 = 4a$$

- (c) Find the acceleration of the system. [0.5]

Ans:

Adding both equations: $m_1g - m_2g = (m_1 + m_2)a$

$$a = (m_1 - m_2)g / (m_1 + m_2) = (6 - 4) \times 9.81 / (6 + 4)$$

$$a = 19.62 / 10 = 1.962 \text{ m/s}^2$$

- (d) Calculate the tension in the string. [0.5]

Ans:

From equation for m_2 : $T = m_2(g + a) = 4 \times (9.81 + 1.962)$

$$T = 4 \times 11.772 = 47.09 \text{ N}$$

Check via m_1 : $T = m_1(g - a) = 6 \times 7.848 = 47.09 \text{ N}$ □

- (e) Find the velocity of the masses after 2 seconds. [0.5]

Ans:

$$v = u + at = 0 + 1.962 \times 2 = 3.924 \text{ m/s}$$

- (f) Calculate the distance traveled by each mass in 2 seconds. [0.5]

Ans:

$$s = \frac{1}{2}at^2 = 0.5 \times 1.962 \times 2^2 = 0.5 \times 1.962 \times 4 = 3.924 \text{ m}$$

Q3 [3 Marks]

Bloom: BL 3-4 (Apply / Analyze) | CO2

A slider-crank mechanism has crank radius $r = 50$ mm, connecting rod length $L = 150$ mm, and operates at 3000 RPM. Analyze the mechanism at crank angle $\theta = 30^\circ$.

- (a) Convert the crank speed from RPM to rad/s. [0.5]

Ans:

$$\omega = 2\pi \times N / 60 = 2\pi \times 3000 / 60 = 314.16 \text{ rad/s}$$

- (b) Calculate the connecting rod angle ϕ using the kinematic relation $r \sin \theta = L \sin \phi$. [0.5]

Ans:

$$\sin \phi = (r/L) \sin \theta = (50/150) \times \sin 30^\circ = 0.3333 \times 0.5 = 0.1667$$

$$\phi = \arcsin(0.1667) = 9.59^\circ$$

- (c) Determine the piston displacement from top dead center (TDC). [0.5]

Ans:

$$\text{Piston position: } x = r \cos \theta + L \cos \phi = 50 \cos 30^\circ + 150 \cos 9.59^\circ$$

$$= 43.30 + 147.90 = 191.20 \text{ mm}$$

$$\text{TDC position: } x_{\text{TDC}} = r + L = 200 \text{ mm}$$

$$\text{Displacement from TDC} = 200 - 191.20 = 8.80 \text{ mm}$$

- (d) Find the piston velocity. [0.5]

Ans:

$$v_{\text{piston}} \approx -r\omega(\sin \theta + (r \sin 2\theta)/(2L))$$

$$= -50 \times 314.16 \times (\sin 30^\circ + (50 \times \sin 60^\circ)/(2 \times 150))$$

$$= -50 \times 314.16 \times (0.5 + 0.1443)$$

$$= -10121 \text{ mm/s} = -10.12 \text{ m/s}$$

- (e) Identify the crank angles at which the piston velocity is zero. [0.5]

Ans:

$v_{\text{piston}} = 0$ at Top Dead Center ($\theta = 0^\circ$) and Bottom Dead Center ($\theta = 180^\circ$), where the crank and connecting rod are collinear.

- (f) State the type of motion executed by the piston and name one engineering application. [0.5]

Ans:

The piston executes reciprocating (oscillatory linear) motion. Application: Internal combustion engines.

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A yo-yo of mass $m = 0.05 \text{ kg}$ has outer radius $R = 0.03 \text{ m}$, axle radius $r = 0.01 \text{ m}$, and moment of inertia $I = 2.25 \times 10^{-5} \text{ kg}\cdot\text{m}^2$ (modeled as a solid disk of radius R). The string unwinds from the axle. Take $g = 9.81 \text{ m/s}^2$.

- (a) Write the translational equation of motion for the yo-yo (vertical forces). **[0.5]**

Ans:

Translation (positive downward): $mg - T = ma$

$$\rightarrow 0.05 \times 9.81 - T = 0.05a \rightarrow 0.4905 - T = 0.05a$$

- (b) Write the rotational equation of motion about the center of mass. **[0.5]**

Ans:

Rotation about CM: $Tr = I\alpha$

$$\rightarrow T \times 0.01 = 2.25\text{e-}05 \times \alpha$$

- (c) State the constraint relating linear acceleration a and angular acceleration α . **[0.5]**

Ans:

String unwinds from the axle of radius r , so: $a = \alpha r$

$$\rightarrow \alpha = a/r = a/0.01 = 100a$$

- (d) Solve for the linear acceleration of the yo-yo. **[0.5]**

Ans:

Substituting $\alpha = a/r$ into the rotational equation:

$$T \times 0.01 = 2.25\text{e-}05 \times (a/0.01) \rightarrow T = 2.25\text{e-}05/0.0001 \times a = 0.225a$$

Substituting into the translational equation:

$$0.4905 - 0.225a = 0.05a$$

$$0.4905 = (0.05 + 0.225)a = 0.275a$$

$$a = 0.4905 / 0.275 = 1.784 \text{ m/s}^2$$

- (e) Calculate the tension in the string. **[0.5]**

Ans:

$$T = m(g - a) = 0.05(9.81 - 1.784) = 0.05 \times 8.026 = 0.4013 \text{ N} \approx 0.401 \text{ N}$$

- (f) Express the acceleration as a fraction of g and explain why the yo-yo descends slower than free fall. **[0.5]**

Ans:

$$a/g = 1.784 / 9.81 = 0.182 \approx 18.2\% \text{ of free fall.}$$

The yo-yo descends slower than free fall because its rotational inertia ($I = 2.25\text{e-}05 \text{ kg}\cdot\text{m}^2$) resists angular acceleration. Part of the gravitational potential energy is converted to rotational kinetic energy, leaving less for translational kinetic energy.

— End of Set 3 Model Answers —

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1, CO2

Answer the following:

- (a) Define a holonomic constraint and give one example from a mechanical system. [0.5]

Ans:

A holonomic constraint is one that can be expressed as an equation relating only the coordinates (and possibly time), not velocities: $f(q_1, q_2, \dots, t) = 0$. Example: A bead constrained to move on a circular wire of radius R satisfies $x^2 + y^2 = R^2$ (holonomic). This reduces the degrees of freedom by one.

- (b) What is the physical significance of the natural frequency of a vibrating system? [0.5]

Ans:

The natural frequency $\omega_n = \sqrt{k/m}$ is the frequency at which the undamped system oscillates freely when displaced from equilibrium. It depends only on system properties (stiffness and mass), not on initial conditions. At resonance, when an external forcing frequency $\omega \approx \omega_n$, the response amplitude becomes very large — this is critical in vibration engineering design.

- (c) Define the centripetal and Coriolis components of acceleration in a rotating reference frame. [0.5]

Ans:

In a frame rotating at angular velocity Ω : • Centripetal acceleration = $\Omega \times (\Omega \times r)$, magnitude $\Omega^2 r$, directed radially inward toward the rotation axis. • Coriolis acceleration = $2\Omega \times v_{\text{rel}}$, magnitude $2\Omega v_{\text{rel}}$, perpendicular to both the rotation axis and the relative velocity. It arises only when the object moves within the rotating frame.

- (d) What is a physical (compound) pendulum? How does it differ from a simple pendulum? [0.5]

Ans:

A physical (compound) pendulum is an extended rigid body (e.g. a rod, disk) that oscillates about a fixed pivot point. Unlike a simple pendulum (point mass on a massless string, $T = 2\pi\sqrt{L/g}$), the physical pendulum's period depends on the moment of inertia about the pivot and the distance from pivot to CM: $T = 2\pi\sqrt{I_{\text{pivot}} / (mgd)}$.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A slider-crank has crank radius $r = 50$ mm and connecting rod length $L = 150$ mm. State the ratio L/r and explain its significance in mechanism design. [0.5]

Ans:

$L/r = 150/50 = 3$. This ratio determines how closely the piston motion approximates simple harmonic motion. Higher L/r ratios (typically 3–5) reduce higher-order harmonics in the piston displacement, making the motion smoother and reducing vibrations.

- (b) For a spring-mass system with $m = 2$ kg and $k = 200$ N/m, calculate the natural frequency ω_n in rad/s. [0.5]

Ans:

$$\omega_n = \sqrt{k/m} = \sqrt{(200/2)} = \sqrt{100} = 10 \text{ rad/s.}$$

- (c) Give an example of a constraint force and explain why it does no work in the direction of the constraint. [0.5]

Ans:

Example: The normal force N on a block resting on a surface. The block is constrained not to penetrate the surface. The normal force acts perpendicular to the surface, and since the block cannot move in that direction (the direction of the constraint), the work done by N is zero: $W = N \cdot \text{displacement}_{\perp} = 0$.

- (d) State two applications of gear trains in mechatronics systems. [0.5]

Ans:

(1) Robotic arm joints — gear reducers provide high torque at low speed for precise positioning. (2) CNC spindle drives — gear trains match motor speed to required cutting speed. Other examples: conveyor belt drives, stepper motor micro-stepping gearboxes.

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO2

Two masses $m_1 = 10$ kg and $m_2 = 6$ kg are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s².

- (a) Draw the free-body diagram for each mass and write the equation of motion for m_1 . [0.5]

Ans:

FBD for m_1 : Weight $m_1g = 10 \times 9.81 = 98.10$ N downward, Tension T upward.

Equation of motion (taking downward positive for m_1): $m_1g - T = m_1a$

$$\rightarrow 98.10 - T = 10a$$

- (b) Write the equation of motion for m_2 . [0.5]

Ans:

FBD for m_2 : Weight $m_2g = 6 \times 9.81 = 58.86$ N downward, Tension T upward.

Equation of motion (taking upward positive for m_2): $T - m_2g = m_2a$

$$\rightarrow T - 58.86 = 6a$$

- (c) Find the acceleration of the system. [0.5]

Ans:

Adding both equations: $m_1g - m_2g = (m_1 + m_2)a$

$$a = (m_1 - m_2)g / (m_1 + m_2) = (10 - 6) \times 9.81 / (10 + 6)$$

$$a = 39.24 / 16 = 2.453 \text{ m/s}^2$$

- (d) Calculate the tension in the string. [0.5]

Ans:

From equation for m_2 : $T = m_2(g + a) = 6 \times (9.81 + 2.453)$

$$T = 6 \times 12.263 = 73.58 \text{ N}$$

Check via m_1 : $T = m_1(g - a) = 10 \times 7.357 = 73.58 \text{ N} \quad \square$

- (e) Find the velocity of the masses after 3 seconds. [0.5]

Ans:

$$v = u + at = 0 + 2.453 \times 3 = 7.357 \text{ m/s}$$

- (f) Calculate the distance traveled by each mass in 3 seconds. [0.5]

Ans:

$$s = \frac{1}{2}at^2 = 0.5 \times 2.453 \times 3^2 = 0.5 \times 2.453 \times 9 = 11.036 \text{ m}$$

Q3 [3 Marks]

Bloom: BL 3–4 (Apply / Analyze) | CO2

A solid sphere of mass $m = 5$ kg and radius $R = 0.2$ m rolls without slipping down an incline of angle $\theta = 30^\circ$. Take $g = 9.81 \text{ m/s}^2$.

- (a) Calculate the moment of inertia of the solid sphere about its center. [0.5]

Ans:

$$I_{\text{cm}} = (2/5)mR^2 = (2/5) \times 5.0 \times 0.2^2 = (2/5) \times 5.0 \times 0.040000000000000001 = 0.080 \text{ kg} \cdot \text{m}^2$$

- (b) State the no-slip constraint relating linear and angular quantities. [0.5]

Ans:

No-slip constraint: $v_{\text{cm}} = \omega R$ and $a_{\text{cm}} = \alpha R$, where ω = angular velocity, α = angular acceleration, R = radius.

- (c) Write the expression for total kinetic energy (translational + rotational) of the rolling sphere. [0.5]

Ans:

$$KE_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}(2/5)mR^2(v/R)^2$$

$$= \frac{1}{2}mv^2 + (1/5)mv^2 = (7/10)mv^2$$

- (d) Find the linear acceleration of the sphere down the incline. [0.5]

Ans:

Newton's second law:

Translation: $mg \sin\theta - f = ma$

Rotation: $fR = I\alpha = (2/5)mR^2(a/R) \rightarrow f = (2/5)ma$

Substituting: $mg \sin\theta - (2/5)ma = ma \rightarrow mg \sin\theta = (7/5)ma$

$$a = (5/7)g \sin\theta = (5/7) \times 9.81 \times \sin 30^\circ = (5/7) \times 4.905 = 3.504 \text{ m/s}^2$$

- (e) Compare this acceleration to that of a frictionless sliding block on the same incline. [0.5]

Ans:

Frictionless sliding: $a_{\text{slide}} = g \sin\theta = 9.81 \times 0.5 = 4.905 \text{ m/s}^2$.

Rolling acceleration (3.504) < sliding acceleration (4.905) because part of the gravitational PE goes into rotational KE.

- (f) Find the velocity of the sphere after rolling 3 m from rest. [0.5]

Ans:

$$v = \sqrt{(2as)} = \sqrt{(2 \times 3.504 \times 3.0)} = \sqrt{21.021} = 4.585 \text{ m/s}$$

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A two-degree-of-freedom spring-mass system has $m_1 = m_2 = 1$ kg, $k_1 = k_3 = 100$ N/m (grounding springs), and coupling spring $k_2 = 50$ N/m connecting the two masses.

- (a) Write the equations of motion for mass m_1 and mass m_2 . [0.5]

Ans:

$$\text{For } m_1: m_1 \ddot{x}_1 + k_1 x_1 + k_2(x_1 - x_2) = 0 \rightarrow \ddot{x}_1 + (k_1 + k_2)x_1 - k_2 x_2 = 0$$

$$\rightarrow \ddot{x}_1 + 150x_1 - 50x_2 = 0$$

$$\text{For } m_2: m_2 \ddot{x}_2 + k_3 x_2 + k_2(x_2 - x_1) = 0 \rightarrow \ddot{x}_2 - k_2 x_1 + (k_2 + k_3)x_2 = 0$$

$$\rightarrow \ddot{x}_2 - 50x_1 + 150x_2 = 0$$

- (b) Express the equations in matrix form $[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$. [0.5]

Ans:

$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\} \text{ where } \{x\} = [x_1, x_2]^T$$

- (c) Write the mass matrix $[M]$ and the stiffness matrix $[K]$. [0.5]

Ans:

$$[M] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ (since } m_1 = m_2 = 1 \text{ kg)}$$

$$[K] = \begin{bmatrix} 150 & -50 \\ -50 & 150 \end{bmatrix} \text{ (N/m)}$$

- (d) Set up the characteristic equation $\det([K] - \omega^2[M]) = 0$ and expand it. [0.5]

Ans:

$$\det([K] - \omega^2[M]) = \det(\begin{bmatrix} 150 - \omega^2 & -50 \\ -50 & 150 - \omega^2 \end{bmatrix}) = 0$$

$$(150 - \omega^2)(150 - \omega^2) - (-50)^2 = 0$$

$$(150 - \omega^2)^2 - 2500 = 0 \text{ (since } K_{11} = K_{22} = 150)$$

- (e) Solve for the two natural frequencies ω_1 and ω_2 . [0.5]

Ans:

$$(150 - \omega^2)^2 = 2500$$

$$150 - \omega^2 = \pm 50$$

$$\omega^2 = 150 - 50 = 100 \text{ or } \omega^2 = 150 + 50 = 200$$

$$\omega_1 = \sqrt{100} = 10.00 \text{ rad/s}$$

$$\omega_2 = \sqrt{200} = 14.14 \text{ rad/s}$$

- (f) Describe the physical meaning of the two mode shapes. [0.5]

Ans:

Mode 1 ($\omega_1 = 10.00$ rad/s): Substituting $\omega_1^2 = 100$ into $(150 - 100)X_1 + (-50)X_2 = 0 \rightarrow 50X_1 = 50X_2 \rightarrow X_1 = X_2$.

Both masses move in the SAME direction (in-phase). The coupling spring is not stretched.

Mode 2 ($\omega_2 = 14.14$ rad/s): $(150 - 200)X_1 + (-50)X_2 = 0 \rightarrow -50X_1 = 50X_2 \rightarrow X_1 = -X_2$. Masses move in OPPOSITE directions (out-of-phase). The coupling spring is maximally compressed/extended, giving a higher frequency.

— End of Set 4 Model Answers —

ANSWER KEY — All Sets

Q No.	Set 1	Set 2	Set 3	Set 4
Q1a (i)	The Coriolis acceleration is the term $2\Omega \times \mathbf{v}_{\text{rel}}$ that arises when an object moves	The parallel axis theorem states that the moment of inertia about any axis parallel to the centroidal axis is $I = I_{\text{cm}} + Md^2$	Rolling without slipping means the contact point of the rolling body has zero velocity	R^2 (holonomic). This reduces the degrees of freedom by one.
Q1a (ii)	An Atwood machine consists of two masses connected by a light inextensible string	The gear ratio (GR) is the ratio of the number of teeth (or radius) of the drive gear to the number of teeth (or radius) of the driven gear	A trapezoidal velocity profile has three phases: (1) constant acceleration (velocity increases linearly), (2) constant velocity, (3) constant deceleration (velocity decreases linearly)	The natural frequency $\omega_n = \sqrt{k/m}$ is the frequency at which the undamped system oscillates
Q1a (iii)	$\int r^2 dm$ (continuous). SI unit: $\text{kg}\cdot\text{m}^2$.	A slider-crank mechanism converts rotary motion (crank) into reciprocating linear motion	The logarithmic decrement δ is the natural log of the ratio of two successive peak amplitudes	In a frame rotating at angular velocity Ω : • Centripetal acceleration = $\Omega \times (\Omega \times \mathbf{r})$
Q1a (iv)	Undamped free vibration: the system oscillates indefinitely at its natural frequency	Critical damping is the boundary between oscillatory (underdamped) and non-oscillatory (overdamped) behavior	In an ideal block-and-tackle system, the mechanical advantage (MA) equals the number of rope segments supporting the load	$2\pi\sqrt{I_{\text{pivot}} / (mgd)}$.
Q1b (i)	$0.30 \text{ kg}\cdot\text{m}^2$.	6.67 RPM.	12 m/s^2 . Direction: perpendicular to $\mathbf{v}_{\text{rel}}$, opposing the sense of rotation.	$L/r = 150/50 = 3$. This ratio determines how closely the piston motion approximates simple harmonic motion
Q1b (ii)	The rolling acceleration is $a = g \sin\theta / (1 + \beta)$, where $\beta = I/(mR^2)$. Solid sphere	The damping ratio $\zeta = c / c_{\text{cr}}$ is the ratio of actual damping to critical damping	restoring spring force.	10 rad/s.
Q1b (iii)	The 5 kg mass accelerates downward because it is heavier. In an Atwood machine,	2.2 s.	Mechanical advantage (MA) = Load / Effort = W / F . For an ideal single movable pulley, MA = 2	0.
Q1b (iv)	(1) The pulley is massless and frictionless (no rotational inertia, no axle friction)	The block remains stationary when the component of gravity along the incline is balanced by static friction	Newton's laws are valid only in inertial (non-accelerating) frames. When we write equations in a non-inertial frame, we must include fictitious forces	(1) Robotic arm joints — gear reducers provide high torque at low speed for precision
Q2 (a)	5a	8a	6a	10a
Q2 (b)	3a	5a	4a	6a
Q2 (c)	2.453 m/s^2	2.264 m/s^2	1.962 m/s^2	2.453 m/s^2
Q2 (d)	36.79 N	60.37 N	47.09 N	73.58 N
Q2 (e)	4.905 m/s	6.792 m/s	3.924 m/s	7.357 m/s
Q2 (f)	4.905 m	10.187 m	3.924 m	11.036 m
Q3 (a)	18.0 m/s^2	$\mu_k N$ opposing motion (up the incline).	314.16 rad/s	$0.080 \text{ kg}\cdot\text{m}^2$
Q3 (b)	9.0 m/s^2	42.48 N	9.59°	No-slip constraint: $\mathbf{v}_{\text{cm}} = \omega R$ and $\mathbf{a}_{\text{cm}} = \alpha R$, where ω = angular velocity, α = angular acceleration
Q3 (c)	The Coriolis acceleration is perpendicular to $\mathbf{v}_{\text{rel}}$ (the radial direction), directed tangentially	8.50 N	8.80 mm	$(7/10)mv^2$
Q3 (d)	20.12 m/s^2	3.21 m/s^2	-10.12 m/s	3.504 m/s^2
Q3 (e)	6.0 m/s	5.66 m/s	180° , where the crank and connecting rod are collinear.	4.905 m/s^2 .
Q3 (f)	36.0 m/s^2	1.77 s	The piston executes reciprocating (oscillatory linear) motion. Application: Internal combustion engine	4.585 m/s
Q4 (a)	$0.6667 \text{ kg}\cdot\text{m}^2$	0	$0.05a$	0
Q4 (b)	0.5 m	$0.6283 \text{ s} \approx 0.628 \text{ s}$	$2.25e-05 \times \alpha$	$[x_1, x_2]^T$
Q4 (c)	0	$0.1 \cos(10t) \text{ m}$	100a	$[[150, -50], [-50, 150]] \text{ (N/m)}$
Q4 (d)	3.836 rad/s	0.10	1.784 m/s^2	150)
Q4 (e)	1.638 s	9.95 rad/s	$0.4013 \text{ N} \approx 0.401 \text{ N}$	14.14 rad/s
Q4 (f)	2.006 s .	0.5318 times the previous peak.	$0.182 \approx 18.2\%$ of free fall.	Mode 2 ($\omega_2 = 14.14 \text{ rad/s}$): $(150-200)X_1 + (-50)X_2 = 0 \rightarrow -50X_1 = 50X_2 \rightarrow X_1 = -X_2$.